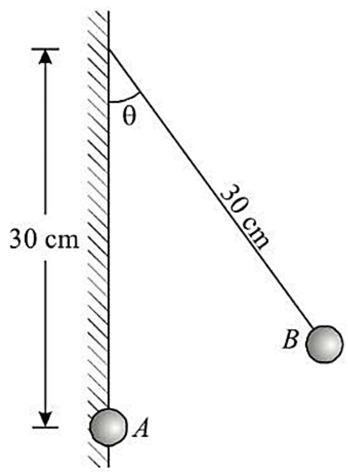
Lakshya JEE 2027

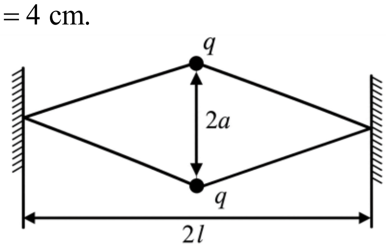
KPP – 07 **Electrostatics** **By: Saleem Bhaiya**

1. A ball of mass m with a charge q can rotate in a vertical plane at the end of a string of length l in a uniform electrostatic field whose lines of force are directed upwards. What horizontal velocity must be imparted to the ball in the upper position so that the tension in the string in the lower position of the ball is 15 times than the weight of the ball?

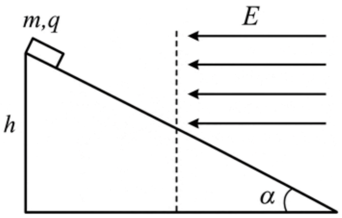
2. A particle A having with a charge $q = 5\times10^{-7}\,C$ is clamped in a vertical wall. A small ball B of mass 100 g and having equal charge is suspended by an insulating thread of length 30 cm from the wall. The point of suspension is 30 cm above the particle A as shown in figure. Find the angle θ which the thread makes with the wall in equilibrium. Take $g = 10\ m/s^2$.



3. Two identical charges q are connected by rubber cords to the walls, as shown in figure at a distance $2a$ from each other. The distance between the walls is $2l$ and the length of each non-deformed cord is l. Determine the force constant of the cord. Mass of charges is negligible. $q = 1\,\mu C$, $a = 3$ cm, $l = 4$ cm.



4. A small object of mass m and of charge q slides down frictionlessly along a slope of angle of elevation α from a height of h. From the midpoint of the slope, it enters into a region of uniform horizontal electric field, and it decelerates and stops at the bottom of the slope.



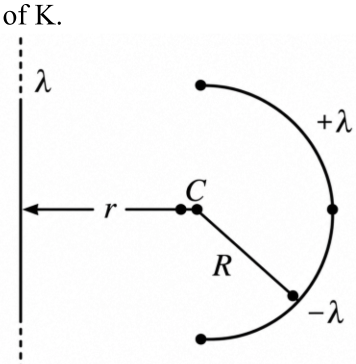
(1) The magnitude of E is $\frac{mg}{q}\tan\alpha$.

(2) The normal reaction on block before dotted line and after dotted line remains the same.

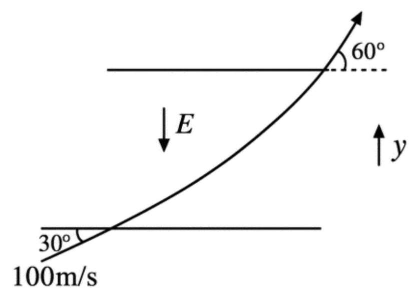
(3) The magnitude of acceleration before and after dotted line remains the same.

(4) The speed at height $\frac{h}{4}$ from bottom is $\sqrt{\frac{gh}{2}}$.

5. In the figure shown, a very long wire and a semicircular ring of radius 'R' are placed in the same plane. The center of the ring is at a distance 'r' from the wire. The wire has uniformly distributed line charge density 'λ' and the ring has linear charge density '$+\lambda$' on one half and '$-\lambda$' on the other half as shown. If $\lambda = 2$C/m, , R = 3 m and r = 1 m, then magnitude of net torque on the ring due to the wire is $\frac{K}{\pi\varepsilon_0}\ln 2\ \text{N} - \text{m}$. Find the value of K.



6. Find the magnitude of uniform electric field E in N/C (direction shown in figure) if an electron entering with velocity 100 m/s making 30° with horizontal comes out making 60° with horizontal (see figure), after a time numerically equal to $\frac{m}{e}$ for electron.



7. A particle is thrown vertically upward from ground level with a speed of $5\sqrt{5}\,\text{m/s}$ in a region of space having uniform electric field. As a result, it attains a maximum height h. The particle is then given a positive charge q and it reaches the same maximum height h when thrown vertically upward with a speed of 13 m/s. Next, the particle is given a negative charge q. Ignoring air resistance, determine the speed with which the negatively charged particle must be thrown vertically upward so that it attains the same maximum height h.

8. A disc of radius R is charged uniformly with charge/area σ. Choose the correct options for electric field at a point on the axis of the disc.

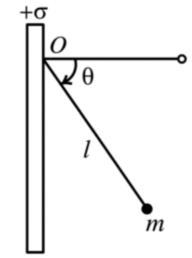
(1) E is maximum at $\frac{R}{\sqrt{2}}$ from the disc.

(2) E just close to the disc is $\frac{\sigma}{2\epsilon_0}$

(3) E decreases continuously as one goes away from the disc.

(4) E at a distance of R from disc is $\frac{\sigma}{4\epsilon_0}$

9. A bob of mass m and charge q is suspended from a point O of a vertical large thin sheet of surface charge density $+\sigma$. If the bob is released from rest from horizontal, at which angle θ the speed of the bob will be (i) maximum (ii) zero.



(1) (i) $\theta_1 = \tan^{-1}\left(\frac{\varepsilon_0 mg}{\sigma q}\right)$

(ii) $\theta_2 = 2\tan^{-1}\left(\frac{\varepsilon_0 mg}{\sigma q}\right)$

(2) (i) $\theta_1 = \tan^{-1}\left(\frac{2\varepsilon_0 mg}{\sigma q}\right)$

(ii) $\theta_2 = 2\tan^{-1}\left(\frac{2\varepsilon_0 mg}{\sigma q}\right)$

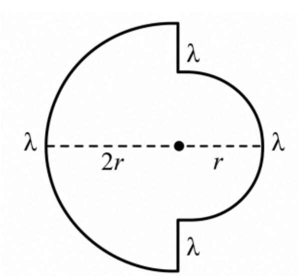
(3) (i) $\theta_1 = 2\tan^{-1}\left(\frac{\varepsilon_0 mg}{\sigma q}\right)$

(ii) $\theta_2 = \tan^{-1}\left(\frac{\varepsilon_0 mg}{\sigma q}\right)$

(4) (i) $\theta_1 = 2\tan^{-1}\left(\frac{2\varepsilon_0 mg}{\sigma q}\right)$

(ii) $\theta_2 = \tan^{-1}\left(\frac{2\varepsilon_0 mg}{\sigma q}\right)$

10. Two semicircular rings lying in same plane, of uniform linear charge density λ have radius r and $2r$. They are joined using two straight uniformly charged wires of linear charge density λ and length r as shown in figure. Find the magnitude of electric field at common center of semi circular rings.



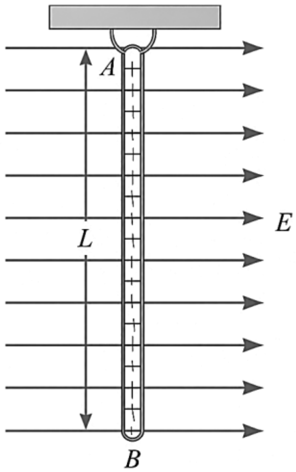
(1) $\frac{\lambda}{4\pi\varepsilon_0 r}$

(2) $\frac{\lambda}{2\pi\varepsilon_0 r}$

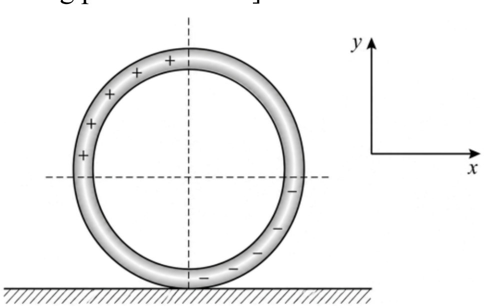
(3) $\frac{\lambda}{\pi\varepsilon_0 r}$

(4) $\frac{2\lambda}{\pi\varepsilon_0 r}$

11. A rod AB of length L and mass m is uniformly charged with a charge Q, and it is suspended from end A as shown in figure. The rod can freely rotate about A in the plane of the figure. An electric field E is suddenly switched on in the horizontal direction due to which the rod gets turned by a maximum angle of 90°. The magnitude of E is equal to nMg/Q. Find the value of n.

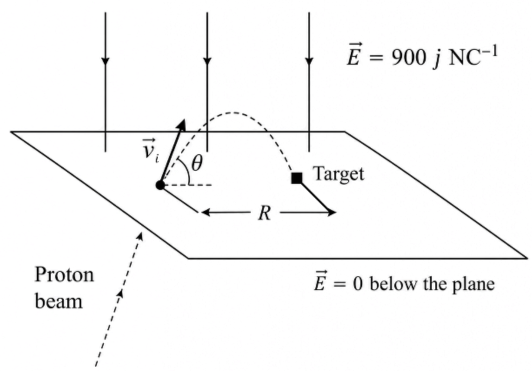


12. A non-conducting ring of mass m and radius R is charged as shown in figure and placed on a rough horizontal non conducting plane. The charge per unit length on the charged quadrants of ring is λ. At time $t = 0$, a uniform electric field $\vec{E} = E_0\hat{i}$ is switched on and the ring starts rolling without sliding. Determine the magnitude and direction of friction force acting on the ring when it starts rolling.
[$\lambda R E_0$ along positive x-axis]



13. Electric field E = $-bx + a$ exists in a region parallel to the x-direction (a and b are positive constants). A charge particle having charge q and mass m is released from the origin $x = 0$. Find the acceleration of the particle at the instant its speed becomes zero for the first time after release.

14. Protons are projected with an initial speed v_i = 6 kms^{-1} from a field-free region $\vec{E} = -900\hat{j}\text{NC}^{-1}$ present above the plane as shown in figure. The initial velocity vector of the protons makes an angle u with the plane. The protons are to hit a target that lies at a horizontal distance of R = 2 mm from the point where the protons cross the plane and enter the electric field. Find the angle θ at which the protons must pass through the plane to strike the target.



Answer Key

1. $\left[\frac{5l}{m}(qE+2mg)\right]^{1/2}$
2. (17°)
3. (208.3)
4. (1, 4)
5. (24)
6. (100)
7. 9 m/s
8. (2, 3)
9. (2)
10. (1)
11. (1)
12. $\lambda RE_0\hat{i}$
13. $-\frac{qa}{m}$
14. (15°)